

Erdős Problem #600

EP-600

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $e(n, r)$ be minimal such that every graph on n vertices with at least $e(n, r)$ edges, each edge contained in at least one triangle, must have an edge contained in at least r triangles. Let $r \geq 2$. Is it true that $e(n, r+1) - e(n, r) \rightarrow \infty$ as $n \rightarrow \infty$? Is it true that $\frac{e(n, r+1)}{e(n, r)} \rightarrow 1$ as $n \rightarrow \infty$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>